



ASHESI UNIVERSITY

INTRODUCTION TO ARTIFICIAL INTELLIGENCE END OF SEM PROJECT AI USE DECLARATION FORM

Cohort B

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AI Use Summary

Question	Team Response
Did you use any AI tool for this lab?	Yes
Estimated percentage of the work influenced by AI	23%
Did you attach evidence of AI use where applicable?	Yes

2. Details of AI Use

Complete the table below for each AI tool used. Add more rows where necessary.

Name of AI Tool Used	Purpose for Using the Tool	Prompt or Instruction Given to the Tool	Part(s) of the Work Influenced by the Tool	How We Verified, Edited, or Improved the AI Output
Gemini	Calculating the exact number of parameters for a custom convolutional neural network (CNN) architecture.	"Formulate an exact mathematical parameter calculation breakdown for a 3-block convolutional network transitioning to a 256-node dense classifier."	In the data exploration and cleaning notebook	Code & Arithmetic Verification: Manually verified all mathematical parameter counts (25,785,578 parameters) by writing tensor assertion scripts and PyTorch parameter inspection loops in Colab.
Gemini	Formulating exact analytical layer parameter formulas to benchmark against MobileNetV3.	"How do we architect and regularize a custom 3-block CNN from scratch in PyTorch to classify a 10-class multi-crop dataset while preventing empirical overfitting?"	In training of our personal model	Validated all training metrics, loss values (\$0.239\$), and accuracy rates (\$92.0\%\$) against live execution logs in
Gemini	Curating localized agricultural diagnostic	"Generate an authentic, dialect-accurate Ghanaian agricultural	Dialect-specific agricultural translation dictionaries found	Cross-checked Twi, Ga, and Ewe agricultural phrasing for cultural resonance

	terminology across Twi (Akan), Ga, and Ewe.	diagnostic dictionary translating crop pathologies and actionable IPM remediation steps into Twi (Akan), Ga, and Ewe."	within src/translator.py.	and technical accuracy regarding local pest management techniques.
ChatGPT	Troubleshooting model performance discrepancies between test data and real-world application.	"My CNN performs well on the held-out test dataset but gives incorrect or low-confidence predictions when given new crop images through the frontend. Explain why test performance may differ from performance on unseen real-world images and how I can determine whether the problem comes from the model, image preprocessing, class-label mapping, or poor generalization."	Model evaluation and deployment strategy.	Analyzed the suggested debugging steps and updated data preprocessing pipeline to better align training and production environments.
ChatGPT	Understanding and interpreting classification model performance metrics.	Explain the difference between accuracy, precision, recall, F1-score, macro F1-score, and weighted F1-score in a multi-class crop disease classification model. Why might a model have high overall accuracy but	Performance analysis and report section.	Verified definitions against course materials and contextualized the explanations for the specific disease classes.

		perform poorly on an underrepresented disease class?"		
ChatGPT	Implementing data augmentation to improve model robustness.	What is the purpose of data augmentation, and how would it help with training the pretrained model?	Model training methodology.	Implemented recommended augmentation techniques and validated their impact by comparing model accuracy before and after.
ChatGPT	Optimizing training efficiency using transfer learning.	Explain to me how a pretrained model's layers are frozen for retraining of the classifier	Model architecture implementation.	Reviewed code snippets and confirmed layer freezing by checking parameter gradients during the initial training epoch.
ChatGPT	Integrating the trained model with the frontend application.	I have a trained PyTorch custom CNN model for GreenPulse saved as a .pth file. Help guide me to connect this model to my application so that when a user uploads a crop leaf image, the application preprocesses the image, passes it to the model, and displays the predicted disease and confidence score	Frontend-backend integration code.	Followed the integration guide and tested the end-to-end data flow with sample inputs.
ChatGPT	Troubleshooting communication between the frontend interface	My GreenPulse interface is displaying the upload and prediction	Application connectivity and interface logic.	Debugged the connection paths based on the advice and confirmed

	and the backend model prediction.	sections, but I need help troubleshooting the connection between the uploaded image, the PyTorch model, and the prediction result shown on the screen. Help me identify what needs to be changed while keeping my existing application structure		successful model inference in the live interface.
ChatGPT	Interpreting complex evaluation metrics and the confusion matrix.	Help me understand these evaluation results for my group's GreenPulse custom CNN, including the classification report and confusion matrix. Identify any problems with the evaluation and explain what the accuracy, precision, recall, F1-score, and per-class accuracy mean.	Evaluation and conclusion section.	Validated the interpretations against the actual confusion matrix data and corrected minor discrepancies in the analysis.

3. Attachment of AI Output Evidence

Where required, attach evidence of AI use. Tick all that apply. Add more rows where necessary.

Evidence Type	Attached? (Yes, No, N/A)	File Name
AI-generated draft	No	N/A
Prompt history	No	N/A
Screenshot of AI interaction	No	N/A

Exported AI conversation	No	N/A
AI-generated code snippet	No	N/A
Revised version showing student input	No	N/A

4. Student Declaration

I declare that:

Declaration Statement	Response (Yes/No)
The submitted work is our own work.	Yes
Any use of AI tools has been clearly declared in this form.	Yes
The prompts, instructions, and AI-generated outputs have been disclosed where applicable.	Yes
We reviewed, tested, edited, and improved any AI-generated content before submission.	Yes
Our AI usage does not exceed 25% of the entire work.	Yes
We understand that undeclared or excessive AI use may be treated as academic misconduct.	Yes

Signatures:

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